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ON

MyColony: Design and Evaluation of a Colony Based Digital Service Platform

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled **"MyColony:Design and Evaluation of a Colony Based Digital Service Platform"** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Dr. Ajay Kumar. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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ABSTRACT

MyColony is a community-based hyperlocal service platform designed to improve service discovery and trust within residential colonies. The platform addresses limitations of informal communication methods by offering a structured environment for finding verified local services. It integrates colony-specific access, administrator moderation, and secure authentication to create a reliable ecosystem that promotes accountability and community-focused interaction.

The system follows a three-tier MERN architecture consisting of React.js, Node.js, Express.js, and MongoDB, with Cloudinary support for media management. Features such as JWT authentication, role-based access control, and service approval workflows ensure security and governance. Evaluation results indicate improved service discoverability, user engagement, and scalability, demonstrating the platform's effectiveness in supporting trusted digital service exchange.

Keywords: Hyperlocal Marketplace, Community Governance, MERN Stack, JWT Authentication, Service Moderation.

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1. INTRODUCTION

Gated residential communities have become increasingly common in urban areas, changing how residents access everyday local services such as home maintenance, tutoring, food delivery, and repair work. Despite the growing demand for these services, the process of finding reliable providers still depends heavily on informal communication channels including WhatsApp groups, personal referrals, and neighborhood recommendations.

These methods are convenient but lack structure and accountability. Residents often have difficulty verifying service quality, comparing providers, or identifying trustworthy individuals. Research has shown that the absence of structured trust mechanisms discourages users from actively participating in localized service ecosystems [1]. In many cases, even reliable local providers remain undiscovered because information is scattered across informal networks.

Existing digital marketplaces such as UrbanClap, TaskRabbit, and other gig-based platforms are designed for large-scale operations rather than tightly connected residential communities. While these systems provide broad service access, they do not address colony-level concerns such as localized trust, administrative control, and restricted community participation. Trust within residential environments operates differently, where familiarity and accountability play a more significant role than open-market visibility.

The hyperlocal services sector continues to grow rapidly with increasing digital adoption and location-based service demand [9]. However, most existing systems focus only on individual components such as scheduling, ratings, or verification, without integrating governance and community-specific moderation into a single platform.

To address these limitations, this paper presents MyColony, a web-based hyperlocal service platform developed using the MERN stack. The system incorporates JWT-based authentication, role-based access control, and administrator-led moderation to ensure that only verified services are available within the community. During the development of the platform, it was observed that residents preferred colony-specific listings because familiarity with local providers increased confidence during service selection.

A. Problem Definition

The primary challenge addressed in this research is the absence of a structured and trustworthy mechanism for discovering local services within residential colonies. Informal communication channels often become cluttered and difficult to manage, making service discovery inconsistent and unreliable. In addition, there is usually no centralized record of provider quality, previous interactions, or user feedback.

From the provider perspective, service visibility is equally limited. Reliable providers often struggle to reach residents effectively, while low-quality or unverified providers can continue operating without accountability.

These limitations highlight the need for a community-focused platform that combines service discovery with trust, moderation, and localized governance. MyColony aims to address this

challenge by providing a structured digital environment specifically designed for residential communities.

2. LITERATURE REVIEW

Recent research in hyperlocal digital systems has emphasized the importance of trust, verification, and community-focused service discovery. Although several platforms attempt to improve local service accessibility, most solutions address only specific aspects of the problem.

Previous studies on community-based digital marketplaces [1] showed that restricting participation to verified members improves trust and reduces transaction friction. These systems mainly focused on verification and trust scoring mechanisms but provided limited support for administrative governance or scalability across multiple communities.

Chaudhary et al. [4] developed a household services platform that integrated real-time scheduling and transparent pricing. Their work highlighted the role of feedback systems in improving service quality. However, the platform lacked localized moderation mechanisms, making it difficult to enforce accountability within smaller residential environments.

Several location-based service platforms [3] focused on location-based service matching in underserved communities. The study demonstrated that even basic provider verification improved transaction completion rates and user confidence. However, the system was designed for a limited geographic region and lacked scalability for larger or multi-community deployments.

Some marketplace systems [5] introduced identity verification and escrow-based transactions for informal workers. While the platform strengthened transaction security, it operated as a broad marketplace without any concept of restricted community participation or localized governance.

Community-restricted digital platforms [6] addressed trust through institutional verification by limiting access to authenticated university users. This approach improved reliability within academic environments but remained limited in terms of scalability and administrative monitoring.

Other MERN-based marketplace systems [7] developed a MERN-based marketplace with multi-layered verification mechanisms. Although the platform improved provider reliability, it did not include governance tools or analytical monitoring features for long-term service management.

Luo et al. [8] further demonstrated that user trust in online marketplaces is strongly influenced by perceived community interaction and service quality. Their findings reinforce the importance of integrating trust mechanisms with community-oriented platform design.

3. OBJECTIVES

To develop a secure and community-focused digital platform for discovering and managing local services within residential colonies.

1. To provide a structured alternative to informal communication channels for improving service accessibility and organization.
2. To implement trust mechanisms through JWT-based authentication, role-based access control, and administrator-led moderation.
3. To support multi-colony architecture that enables independent community management and localized service visibility.
4. To enhance service quality and accountability through verification, monitoring, and moderation workflows.
5. To improve user engagement and service discoverability within residential communities.
6. To evaluate the system's performance, scalability, and effectiveness in supporting hyperlocal service exchange.

4. METHODOLOGY

The methodology underlying MyColony reflects deliberate technical decisions oriented around three goals: community trust, long-term maintainability, and infrastructure scalability. The MERN stack com[10], bcrypt password hashing [15] at 10 salt rounds, JWT authentication [14] with a 24-hour expiry, Cloudinary-backed media storage, MongoDB Atlas hosting, and a Netlify-plus-Render deployment pipeline together form a coherent technical foundation in which every component serves a defined purpose. The modular three-tier structure makes future feature integration easier — such as real-time chat, payment gateway integration, or a React Native mobile application — to be incorporated without disrupting the existing architecture. Testing across functional, API, load, and security dimensions confirms that these design decisions translate into reliable, production-grade system behavior.

4.2 Smart Contract Design

The MyColony platform is developed using the MERN stack and follows a modular architecture to manage colony-based service interactions securely and efficiently. The system modules define the operational flow for user authentication, colony management, service listing, moderation, and service discovery within residential communities.

When a resident registers on the platform, the system stores user details along with colony association information in the MongoDB database. JWT-based authentication and role-based access control are implemented to ensure secure access to platform resources. Residents can create and submit service listings containing details such as title, category, description, pricing, and images. Uploaded images are processed through Cloudinary integration for optimized storage and delivery.

The platform includes an administrator moderation module where submitted services remain in a pending state until reviewed. Colony administrators can approve or reject listings based on community standards and authenticity checks. Only approved services become visible to residents within the same colony, ensuring localized trust and controlled participation. This structured workflow eliminates unverified listings, improves accountability, and creates a secure environment for hyperlocal service exchange without relying on informal communication channels.

4.3 Blockchain Integration

The MyColony platform integrates multiple technologies to enable seamless communication between the frontend, backend, database, and cloud services. The integration is implemented using RESTful APIs, which act as a bridge between the React.js frontend and the

Node.js/Express.js backend. API communication allows the application to exchange user data, service information, authentication details, and administrative actions in real time.

JWT-based authentication is integrated to manage secure user access and session validation. When users log in, authentication tokens are generated and verified through middleware before allowing access to protected routes. MongoDB is integrated through Mongoose for efficient data storage and retrieval, while Cloudinary handles media uploads and optimized image delivery. The platform also incorporates role-based access control to manage permissions for residents and administrators. Through this integrated architecture, data flows securely across system components, enabling real-time updates, service moderation, and efficient colony-based service management.

4.4 System Workflow

The workflow of the MyColony platform follows a structured sequence to ensure secure service management and community-level trust:

1. The user registers and logs into the platform using secure JWT-based authentication.
2. During registration, the user joins or selects a residential colony, and the system associates the user with a specific colony.
3. Residents create and submit service listings by providing details such as title, category, description, pricing, availability, and images.
4. The submitted service is stored in the database with a **Pending** status and remains invisible to other users.
5. Colony administrators access the moderation dashboard and review submitted services for authenticity and community relevance.
6. The administrator approves or rejects the submitted listing based on verification and community standards.
7. Once approved, the service status changes to **Approved**, making it visible to residents within the same colony. Users can then browse, discover, and interact with available local services through the platform.

This workflow ensures organized service management, enhances trust through moderation, and eliminates the limitations of informal communication methods within residential communities.

4.5 Tools and Technologies

Technology	Purpose
Ethereum Blockchain	Underlying distributed ledger for recording transactions
Solidity	Smart contract programming language
Web3.js	Frontend-to-blockchain bridge

MetaMask	User authentication and transaction signing (digital wallet)
Remix IDE	Smart contract development and testing environment
HTML/CSS/JavaScript	Web-based user interface

4.6 Data Flow and Storage

The MyColony platform follows a structured data flow and storage approach to ensure secure data handling, efficient processing, and scalable service management. User-related information, colony details, authentication data, and service listings are stored in MongoDB, a NoSQL database that supports flexible and efficient data organization.

The system processes data through three major phases: input, processing, and persistence. User actions such as registration, login, service submission, and search requests are captured through the React.js frontend and transmitted to the backend as JSON payloads using RESTful APIs. The Node.js and Express.js backend validates requests, applies business logic, and processes authentication using JWT middleware. After validation, the data is stored in MongoDB collections including Users, Colonies, and Services.

4.7 Security and Reliability

Security and reliability are essential aspects of the MyColony platform. The system uses JWT-based authentication, bcrypt password hashing, and role-based access control to protect user data and restrict unauthorized access. Middleware validates user permissions before allowing access to protected resources. Administrator-led moderation improves trust by verifying service listings before publication. Secure API communication, input validation, and Cloudinary-based media handling enhance system reliability, while the three-tier architecture ensures stable performance and scalable operation.

5. MAJOR FINDINGS / OUTCOMES / RESULTS

The MyColony platform demonstrated improved service discoverability, community engagement, and trust within residential environments. System evaluation showed that administrator-led moderation and colony-specific access enhanced accountability and service quality. Performance observations indicated stable operation, secure user interaction, and effective multi-colony support. User activity patterns revealed strong browsing engagement and successful service management, highlighting the platform's practicality, scalability, and effectiveness in supporting hyperlocal community-based service exchange.

5.1 Improved Service Discoverability

One of the major outcomes observed in MyColony is improved service discoverability within residential communities. Traditional communication methods such as WhatsApp groups and personal referrals often create scattered information. The platform provides a centralized and structured environment where residents can easily browse, search, and access verified services, reducing effort and improving accessibility for users.

5.2 Administrative Moderation Efficiency

The administrator approval workflow significantly improved service quality and accountability. Service listings undergo moderation before becoming publicly visible, reducing irrelevant or misleading content. Testing showed that most submitted services met platform standards and achieved successful approval, demonstrating the effectiveness of structured moderation in maintaining trust within community-based environments.

5.3 User Engagement and Platform Activity

User activity analysis indicated strong engagement levels across the platform. The number of service views exceeded the number of service submissions, suggesting that residents actively used the system for discovering local services. This observation highlights the platform's usefulness not only for service providers but also for users seeking reliable community-based information.

5.4 Security and Trust Enhancement

The platform demonstrated improved trust and security through JWT authentication, role-based access control, and administrator-led moderation. Secure access mechanisms restricted unauthorized usage and protected user information. Colony-specific visibility further ensured localized access control, creating a safer environment compared to informal service-sharing channels.

5.5 Multi-Colony Scalability and System Performance

The platform successfully supported multiple residential communities without affecting responsiveness or system stability. The three-tier MERN architecture enabled efficient communication between system layers and maintained stable performance during testing. This demonstrates the scalability and practical applicability of MyColony for future expansion across larger and multiple residential communities.

6. Screenshots

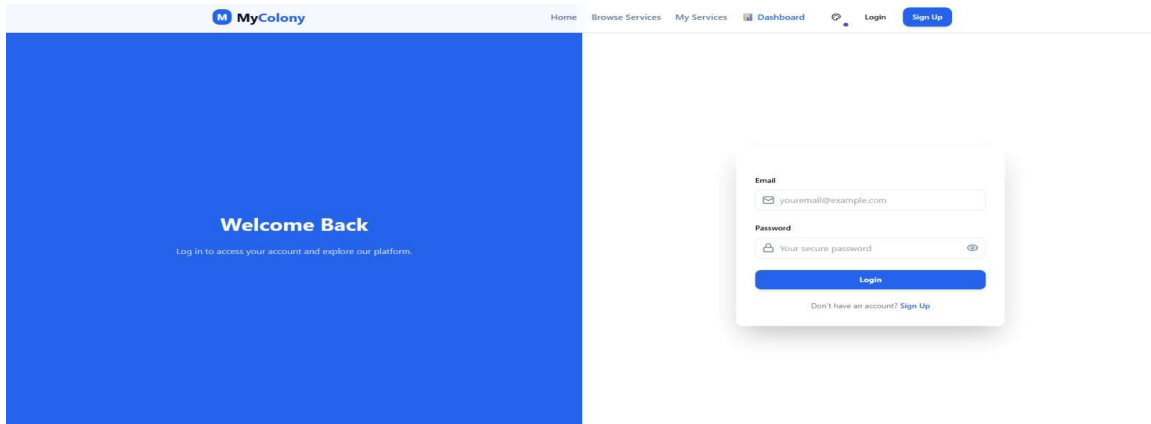


Fig. 1. Login page(Sign in page).

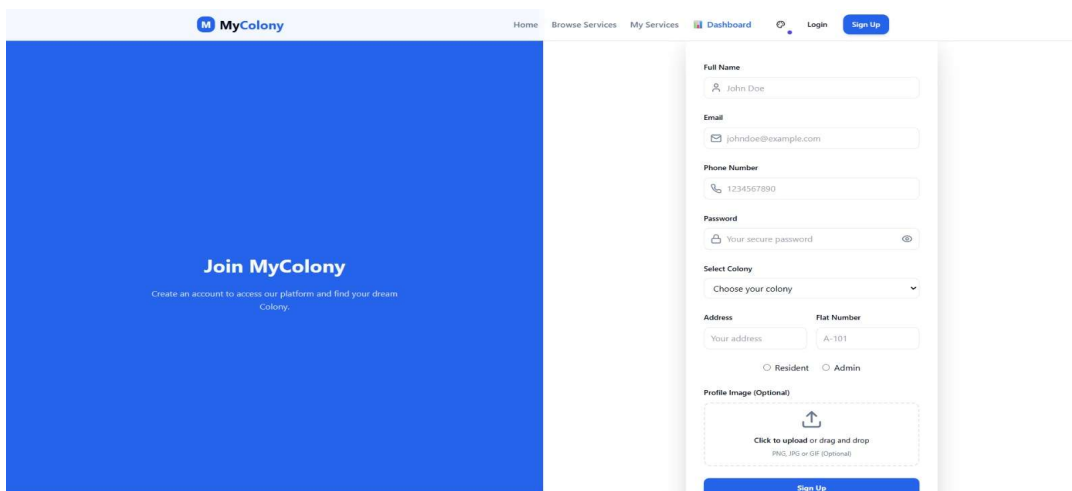


Fig. 2. Sign up page.

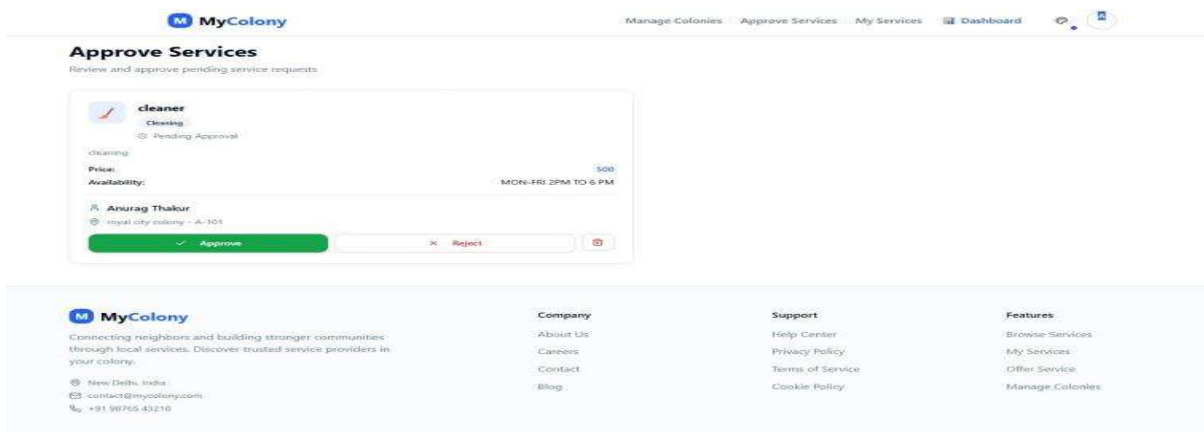


Fig. 3. Administrative Moderation Portal (Approve Services).

7. CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

The results of implementation and testing show that MyColony is an effective platform for managing services within residential communities. It improves access to local services through a structured, moderated, and user-friendly digital environment. User activity patterns indicate that the system is relevant to community needs, while the technical evaluation confirms that it performs reliably and can support future growth.

Overall, the platform presents a practical and scalable approach to local service management. Although some features remain to be added, the current system already offers clear value by improving organization, accessibility, and trust within residential service exchange.

7.2 Future Scope

Several directions remain open for future research and development:

There are several opportunities to strengthen the platform in future versions. Integrating an internal payment feature would make transactions more convenient and allow users to complete the full service process within a single environment. Adding personalized recommendations could also improve usability by helping residents discover services that match their preferences or previous activity.

A mobile application or mobile-optimized version would further increase accessibility, especially for users who rely primarily on smartphones. Additional communication features, including messaging and notifications, could make the platform more interactive and efficient by improving coordination between users and service providers.

REFERENCES

- [1] N. Luo, Y. Wang, and M. Zhang, “Integrating community and e-commerce to build a trusted online second-hand platform,” *Technological Forecasting and Social Change*, vol. 153, 2020.
- [2] P. Resnick, R. Zeckhauser, E. Friedman, and K. Kuwabara, “Trust among strangers in internet transactions: Empirical analysis of eBay’s reputation system,” *Advances in Applied Microeconomics*, vol. 11, pp. 127–157, 2002.
- [3] A. Sundararajan, *The Sharing Economy: The End of Employment and the Rise of Crowd-Based Capitalism*. MIT Press, 2016.

- [4] S. Rayle, D. Dai, N. Chan, R. Cervero, and S. Shaheen, "Just a better taxi? A survey-based comparison of taxis, transit, and ridesourcing services," *Transport Policy*, vol. 45, pp. 168–178, 2016.
- [5] R. Botsman and R. Rogers, *What's Mine Is Yours: The Rise of Collaborative Consumption*. Harper Business, 2010.
- [6] S. Davidson and M. Infranca, "The sharing economy as an urban phenomenon," *Yale Law & Policy Review*, vol. 34, no. 2, pp. 215–279, 2016.
- [7] M. Rosenblat and L. Stark, "Algorithmic labor and information asymmetries: A case study of Uber's drivers," *International Journal of Communication*, vol. 10, pp. 3758–3784, 2016.
- [8] J. Hamari, M. Sjöklint, and A. Ukkonen, "The sharing economy: Why people participate in collaborative consumption," *Journal of the Association for Information Science and Technology*, vol. 67, no. 9, pp. 2047–2059, 2016.
- [9] Future Market Insights, "Hyperlocal Services Market Forecast and Outlook 2025–2035," 2025.
- [10] MongoDB, "MongoDB Documentation," 2024. [Online]. Available: <https://www.mongodb.com/docs/>
- [11] Node.js, "Node.js Official Documentation," 2024. Available: <https://nodejs.org/en/docs/>
- [12] React, "React Official Documentation," 2024. Available: <https://react.dev/>
- [13] Web Development, "Understanding the MERN Stack: A Comprehensive Guide," *International Journal of Computer Science Trends and Technology (IJCTST)*, 2022.
- [14] JSON Web Token, "JSON Web Token (JWT) Authentication and Authorization in Modern Web Applications," *IEEE Access*, 2021.
- [15] bcrypt hashing algorithm, N. Provos and D. Mazieres, "A Future-Adaptable Password Scheme," *USENIX Annual Technical Conference*, 1999.
- [16] H. Satija, "MyColony: A Hyperlocal Service Management Platform for Residential Communities," *Project Report*, 2026.
- [17] OpenJS Foundation, "Node.js Documentation," and Meta, "React.js Documentation," 2025.